Slew-driven

When DP is heavy, a slew check plus mid-edge buffer is a fast practical heuristic

The idea

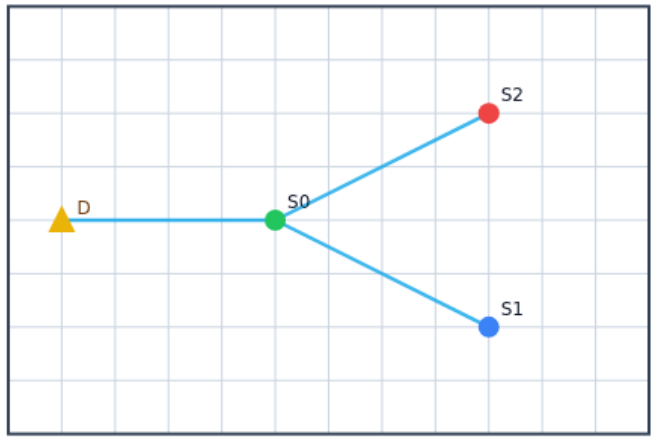
- For each edge compute slew; if above limit, insert BUF_X1 at the midpoint
- Lowering the limit forces more insertions

Check slew

SLEW-DRIVEN

Check slew

Step 1 / 5 · check · module02-07-slew-driven



Explanation

Compare estimate_slew to slew_limit.

- Per edge

```
Gate
```

Check slew

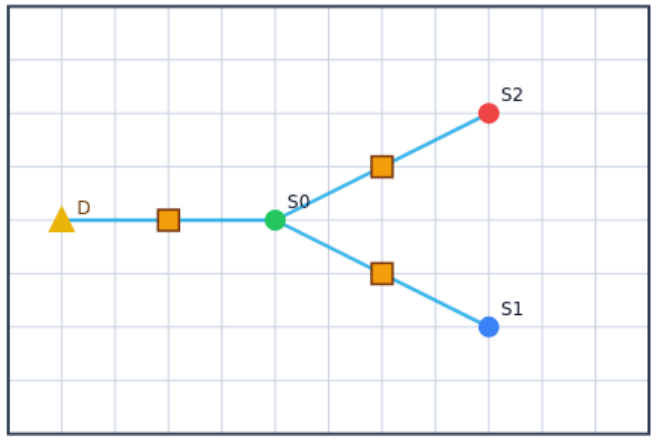
Tight limit

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Tight limit

Step 2 / 5 · tight · module02-07-slew-driven

inserts 3



Explanation

Limit 1 forces inserts on the toy.

- Heuristic

```
count≥1
```

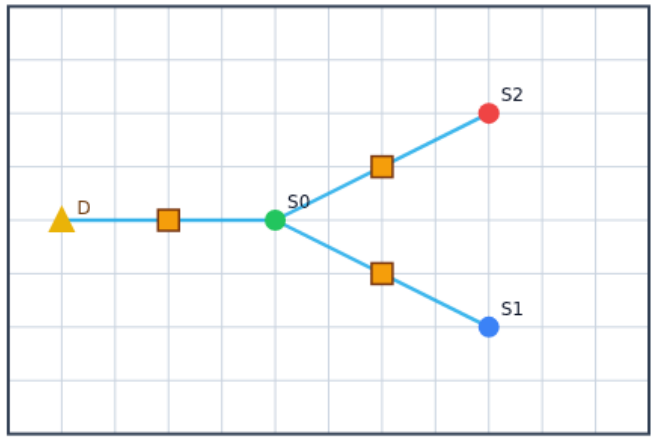
Tight limit

Mid buffer

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Mid buffer

Step 3 / 5 · mid · module02-07-slew-driven



Explanation

Insertion sits at edge midpoint.

- BUF_X1

```
Orange square
```

Mid buffer

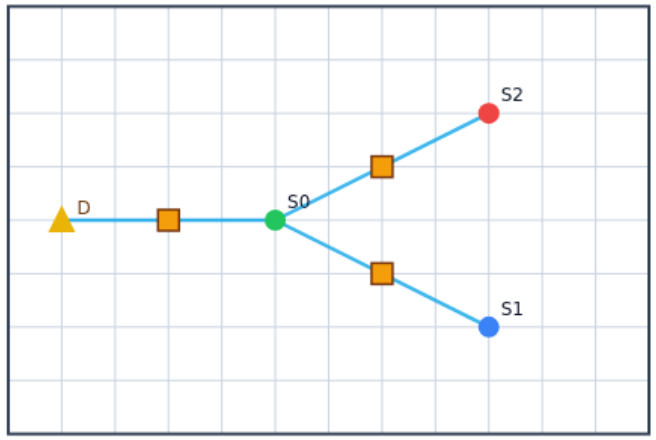
Count

SLEW-DRIVEN

Count

Step 4 / 5 · count · module02-07-slew-driven

count 3



Explanation

Challenges score insertion count.

- Learner state
- Check

```
limit=8
```

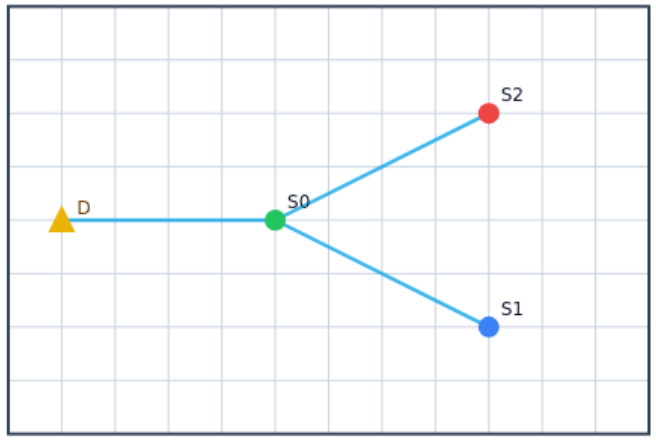
Count

vs VG

SLEW-DRIVEN

vs VG

Step 5 / 5 · vs · module02-07-slew-driven



Explanation

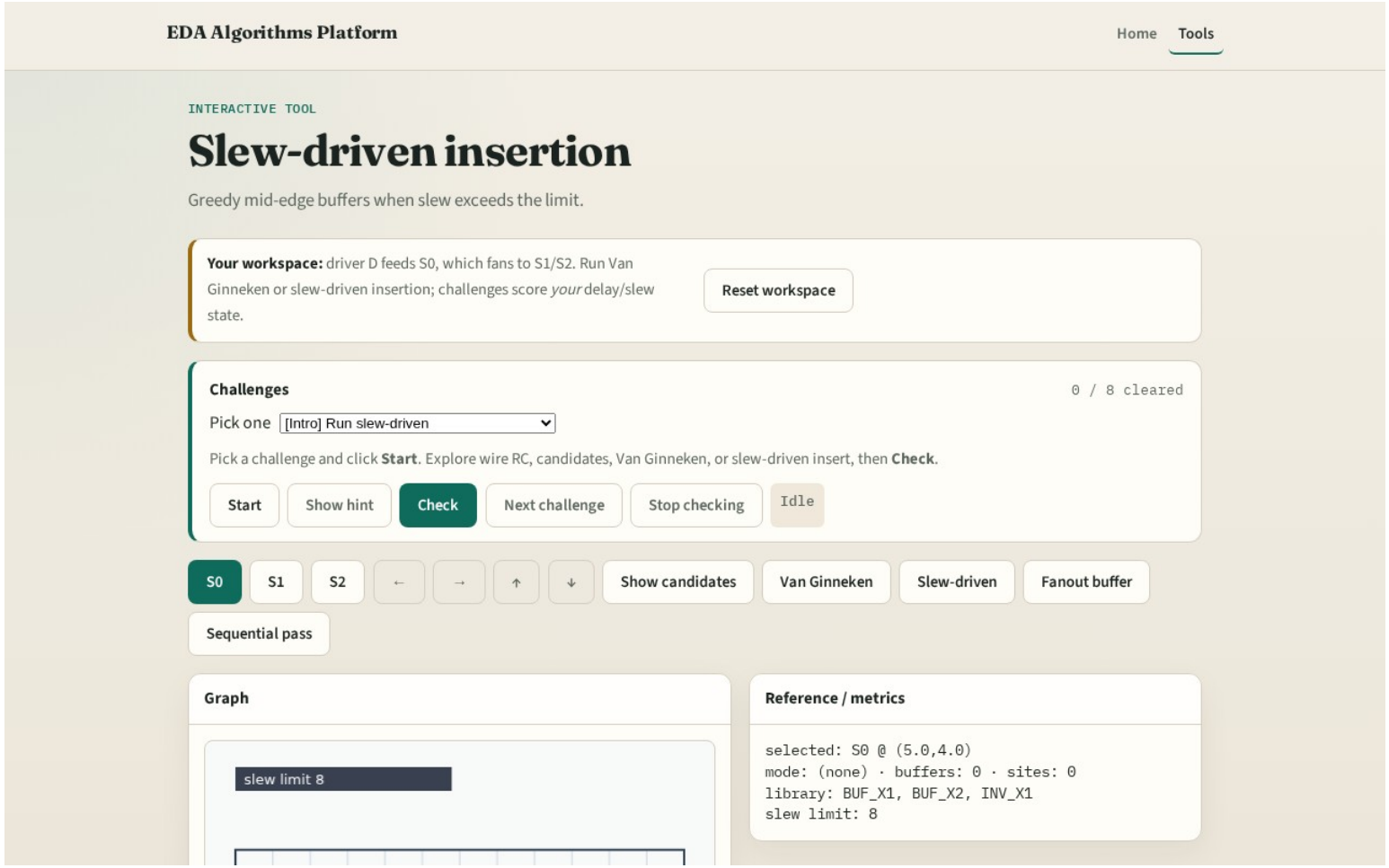
Slew-driven is local; VG optimizes delay options.

- Complementary

```
Both useful
```

vs VG

Browser lab track



Browser lab starter

Implement track

- Call `slew_driven_insert` with slew_limit=1; assert count ≥ 1

Pitfalls

- Inserting without measuring slew
- Stacking infinite buffers
- Ignoring branch load

Your turn

- Next: polarity and inverters